SUMMARY

- When an object or a particle moves back and forth repeatedly about a reference point for some duration of time it is said to have Oscillatory (or vibratory) motion.
- For a SHM, the acceleration or force on the particle is directly proportional to its displacement from a fixed point and always directed towards that fixed point. The force is
$$F_x = -k\,x$$
where k is a constant whose dimension is force per unit length, called as force constant.
- In Simple harmonic motion, the displacement, $y = A \sin \omega t$.
- In Simple harmonic motion, the velocity, $v = \mathrm{A}\,\omega \cos \omega t = \omega\sqrt{A^2 - y^2}$.
- In Simple harmonic motion, the acceleration, $a = \frac{d^2y}{dt^2} = -\omega^2 y$.
- The time period is defined as the time taken by a particle to complete one oscillation. It is usually denoted by T. Time period $T = \frac{2\pi}{\omega}$.
- The number of oscillations produced by the particle per second is called frequency. It is denoted by f. SI unit for frequency is S^{-1} or hertz (In symbol, Hz). Mathematically, frequency is related to time period by $f = \frac{1}{T}$.
- The frequency of the angular harmonic motion is $f = \frac{1}{2\pi}\sqrt{\frac{\kappa}{I}}$ Hz
- For n springs connected in series, the effective spring constant in series is
$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} + \ldots + \frac{1}{k_n} = \sum_{i=0}^{n} \frac{1}{k_i}$$
- For n springs connected in parallel, the effective spring constant is
$$k_p = \sum_{i=1}^{n} k_i$$
- The time period for U-tube oscillation is $T = 2\pi\sqrt{\frac{l}{2g}}$ second.
- For a conservative system in one dimension, the force field can be derived from a scalar potential energy: $F = -\frac{dU}{dx}$.
- In a simple harmonic motion, potential energy is $U(x) = \frac{1}{2}m\omega^2 x^2$.
- In a simple harmonic motion, kinetic energy is $KE = \frac{1}{2}m v_x^2 = \frac{1}{2}m\omega^2(\mathrm{A}^2 - x^2)$.
- Total energy for a simple harmonic motion is $E = \frac{1}{2}m\omega^2 A^2 = \text{constant}$.
- Types of oscillations – Free oscillations, Damped oscillations, Maintained oscillations and Forced oscillations.
- Resonance is a special case of forced oscillations.

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